

Amelia Arabe

SYSTEMS ENGINEER & FOUNDER · EMBEDDED SYSTEMS · HARDWARE-SOFTWARE INTEGRATION

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SUMMARY

I am a Computer and Electrical Design Engineer who builds at the seam of hardware and software. Whether I'm architecting autonomous systems, developing machine learning pipelines, or scaling full-stack platforms, my focus is on creating technology that is intentional and sustainable. As a founder and community leader, I bring a rare blend of technical precision and human-centric strategy to every project. I don't just write code or design circuits; I build systems that solve problems for people.

TECHNICAL SKILLS

HARDWARE & EMBEDDED

FPGA Design (Xilinx / Altera)
Digital Logic & Circuit Design
Microprocessor Systems
Hardware-Software Co-design
Autonomous Systems Design
Biometric & Sensor Integration
Sustainable Systems Design

SOFTWARE & ARCHITECTURE

Embedded C / C++ / Python
JavaScript / GraphQL
Firebase / Supabase / REST APIs
Full-Stack Architecture
Voice Recognition Systems
Object-Oriented & System Design
Machine Learning (fundamentals)

LEADERSHIP & DESIGN

UI/UX Prototyping & Design Systems
Technical Documentation
Brand Strategy & Visual Identity
Cross-functional Team Management
Product Lifecycle Management
Adobe Suite · Figma · Framer
English · Tagalog · German

PROFESSIONAL EXPERIENCE

Ballmecca Software Engineer Intern → Senior Software Engineer & Co-Founder Los Angeles, CA · 2024 – Present

- Co-founded and scaled a software product as founding engineer — architecting a Firebase backend with cross-platform integration that improved system performance by ~40% and supports the full product roadmap.
- Built a reusable component library and design system that reduced development time by ~25% and cut UI bugs by ~50%.
- Integrated APIs and front-end architecture that accelerated feature delivery by ~30%, validated continuously against real user research.

Morenita Design Founder & Software Engineer Temecula, CA · 2025 – Present

- Founded and scaled an independent software and systems studio — directing end-to-end product lifecycles for 7 clients across health & wellness, entertainment, and e-commerce, with 4 active engagements.
- Architect and ship full-stack applications independently — owning system design, database architecture (Supabase), front-end, and client delivery from zero to production.

SELECTED CLIENTS

- **Kairos** — Fertility Tracking & E-Commerce · Built a full-stack fertility tracking application with cycle logging, health analytics, a personalized dashboard, and integrated e-commerce.
- **Professor X Spiritual Guidance Academy** — Platform & E-Commerce · Designed and built a custom platform for a spiritual coaching practice covering course delivery, session booking, community engagement, and e-commerce.
- **YK Entertainment** — B2B SaaS · Architected a platform connecting Southern California vendors and event planners with booking, scheduling, and client management tools.
- **Grupo RPM Health** — Graphic Design · Produced graphic design for PPE product lines including marketing materials, packaging, and brand assets.

Voice Activated Hospital Bed

Baltimore, MD · 2025

Systems Design & Project Lead · Volunteers for Medical Engineering

- Led a 4-person cross-functional team as Systems Designer and Project Manager to engineer a low-cost, voice-controlled hospital bed for quadriplegic patients — integrating real-time voice recognition, biometric monitoring, and safety-critical command validation into a single embedded system.
- Delivered a clinically validated prototype through Volunteers for Medical Engineering, taking the system from schematic to functional deployment.

SLF Terminator Rover Autonomous Systems Engineering · Loyola Arboretum

Baltimore, MD · 2024

- Served as Systems Architect and Team Lead for an autonomous rover designed to combat invasive Spotted Lantern Flies — directing hardware-software co-design across motor control, sensor arrays, and embedded decision logic for real-time detection and path navigation.
- Completed field validation at the Loyola Arboretum with confirmed path following, obstacle avoidance, and live insect attraction — delivering a fully functional autonomous system end-to-end.

Crown HQ Personal Platform OS · Miss Temecula Valley USA 2026

2026

- Architected a 17-panel personal campaign OS — treating a real-world operational goal as a product engineering problem, designing for 4 concurrent users with Kanban management, milestone tracking, file sharing, advocacy tooling, and a live campaign page.
- Backed by Supabase with real-time sync; supports full state-level pageant operations as a candidate for Miss California USA 2026.

EDUCATION

Loyola University Maryland

B.S. Computer Engineering & Electrical Engineering · Class of 2025

Baltimore, MD

FPGA Design · VLSI Design · Digital Logic · Computer Architecture · Microprocessing Systems · Linear Circuit Analysis · Biomedical Imaging & Sensing · Electromagnetics · Quantum Computing · Object-Oriented Design · Experimental Methods

Campus Leadership

Society of Women Engineers — President (Sep 2023 – May 2025) · Oversaw strategy and operations for a 100+ member engineering chapter — directing a leadership board of 10 officers, executing 15+ annual professional development and STEM outreach events, and building industry and academic partnerships to expand access for women in engineering.

BSA Fashion Show — Scene Leader, 25th Annual (2023 – 2024) · Directed choreography, lighting, sound, and media production for 20 models within a 100-model production.

LEADERSHIP & COMMUNITY

Miss Temecula Valley USA 2026

2026

- Leverage a public platform to advocate for sustainable technology and green energy access — presenting technical concepts to community stakeholders and local officials.

Direct a full campaign operation including team coordination, public relations, and platform strategy as a candidate for Miss California USA 2026.

Miss Philippines USA

Los Angeles, CA · 2025

Top Model Award

- Won Top Model Award — voted by photographers for best runway walk and best to work with on set. Selected as Bontika Swim ambassador for sustainable fashion advocacy.

Temecula Valley Youth Orchestra

Temecula, CA · 2018 – 2021

Lead Double Bassist

- Served as Lead Double Bassist — an achievement that earned a performance seat alongside the adult Temecula Valley Orchestra ensemble, one of the few youth members selected to do so.

Cadenza String Orchestra Cellist & Double Bassist

2017 – 2019

- Performed cello and double bass; the orchestra took 1st place at the 2019 Summa Cum Laude International Youth Music Festival in Vienna, Austria.